Class 08: Breast Cancer Mini Project

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Background

The goal of this mini-project is for you to explore a complete analysis using the unsupervised learning techniques covered in class.

Today we will analyze a data-set from a fine needle aspiration (FNA) of a breast mass.

Data Import

The data is made available as a CSV file for download. We can read this in using `read.csv()`.

```r
# Save your input data file into your Project directory
fna.data <- "WisconsinCancer.csv"

# Complete the following code to input the data and store as wisc.df
wisc.df <- read.csv(fna.data, row.names=1)

head(wisc.df)
```

	diagnosis	radius_mean	texture_mean	perimeter_mean	area_mean
842302	M	17.99	10.38	122.80	1001.0
842517	M	20.57	17.77	132.90	1326.0
84300903	M	19.69	21.25	130.00	1203.0
84348301	M	11.42	20.38	77.58	386.1
84358402	M	20.29	14.34	135.10	1297.0
843786	M	12.45	15.70	82.57	477.1

	smoothness_mean	compactness_mean	concavity_mean	concave.points_mean
842302	0.11840	0.27760	0.3001	0.14710
842517	0.08474	0.07864	0.0869	0.07017
84300903	0.10960	0.15990	0.1974	0.12790
84348301	0.14250	0.28390	0.2414	0.10520
84358402	0.10030	0.13280	0.1980	0.10430
843786	0.12780	0.17000	0.1578	0.08089

	symmetry_mean	fractal_dimension_mean	radius_se	texture_se	perimeter_se
842302	0.2419	0.07871	1.0950	0.9053	8.589
842517	0.1812	0.05667	0.5435	0.7339	3.398
84300903	0.2069	0.05999	0.7456	0.7869	4.585
84348301	0.2597	0.09744	0.4956	1.1560	3.445
84358402	0.1809	0.05883	0.7572	0.7813	5.438
843786	0.2087	0.07613	0.3345	0.8902	2.217

	area_se	smoothness_se	compactness_se	concavity_se	concave.points_se
842302	153.40	0.006399	0.04904	0.05373	0.01587
842517	74.08	0.005225	0.01308	0.01860	0.01340
84300903	94.03	0.006150	0.04006	0.03832	0.02058
84348301	27.23	0.009110	0.07458	0.05661	0.01867
84358402	94.44	0.011490	0.02461	0.05688	0.01885
843786	27.19	0.007510	0.03345	0.03672	0.01137

	symmetry_se	fractal_dimension_se	radius_worst	texture_worst
842302	0.03003	0.006193	25.38	17.33
842517	0.01389	0.003532	24.99	23.41
84300903	0.02250	0.004571	23.57	25.53
84348301	0.05963	0.009208	14.91	26.50
84358402	0.01756	0.005115	22.54	16.67
843786	0.02165	0.005082	15.47	23.75

	perimeter_worst	area_worst	smoothness_worst	compactness_worst
842302	184.60	2019.0	0.1622	0.6656
842517	158.80	1956.0	0.1238	0.1866
84300903	152.50	1709.0	0.1444	0.4245
84348301	98.87	567.7	0.2098	0.8663
84358402	152.20	1575.0	0.1374	0.2050
843786	103.40	741.6	0.1791	0.5249

	concavity_worst	concave.points_worst	symmetry_worst

```
842302             0.7119                0.2654          0.4601
842517             0.2416                0.1860          0.2750
84300903           0.4504                0.2430          0.3613
84348301           0.6869                0.2575          0.6638
84358402           0.4000                0.1625          0.2364
843786             0.5355                0.1741          0.3985
         fractal_dimension_worst
842302                   0.11890
842517                   0.08902
84300903                 0.08758
84348301                 0.17300
84358402                 0.07678
843786                   0.12440
```

Make sure we remove or exclude `diagnosis` column from the data-set that we use for further analysis - this is the expert diagnosis as either M or B:

```r
# We can use -1 here to remove the first column
wisc.data <- wisc.df[,-1]
diagnosis <- as.factor(wisc.df$diagnosis)
```

Exploratory Data Analysis

> Q1. How many observations are in this dataset?

```r
nrow(wisc.data)
```

```
[1] 569
```

> Q2. How many of the observations have a malignant diagnosis?

```r
table(wisc.df$diagnosis)
```

```

  B   M 
357 212 
```

> Q3. How many variables/features in the data are suffixed with _mean?

We can use `grep()` function to help us here:

```r
length(grep("_mean", colnames(wisc.data)))
```

```
[1] 10
```

Principal Component Analysis (PCA)

We need to scale our data before PCA with the `scale=TRUE` argument to `prcomp()`

```r
wisc.pr <- prcomp(wisc.data, scale=TRUE)
summary(wisc.pr)
```

```
Importance of components:
                          PC1    PC2     PC3     PC4     PC5     PC6     PC7
Standard deviation     3.6444 2.3857 1.67867 1.40735 1.28403 1.09880 0.82172
Proportion of Variance 0.4427 0.1897 0.09393 0.06602 0.05496 0.04025 0.02251
Cumulative Proportion  0.4427 0.6324 0.72636 0.79239 0.84734 0.88759 0.91010
                           PC8    PC9    PC10   PC11    PC12    PC13    PC14
Standard deviation     0.69037 0.6457 0.59219 0.5421 0.51104 0.49128 0.39624
Proportion of Variance 0.01589 0.0139 0.01169 0.0098 0.00871 0.00805 0.00523
Cumulative Proportion  0.92598 0.9399 0.95157 0.9614 0.97007 0.97812 0.98335
                          PC15    PC16    PC17    PC18    PC19    PC20   PC21
Standard deviation     0.30681 0.28260 0.24372 0.22939 0.22244 0.17652 0.1731
Proportion of Variance 0.00314 0.00266 0.00198 0.00175 0.00165 0.00104 0.0010
Cumulative Proportion  0.98649 0.98915 0.99113 0.99288 0.99453 0.99557 0.9966
                          PC22    PC23   PC24    PC25    PC26    PC27    PC28
Standard deviation     0.16565 0.15602 0.1344 0.12442 0.09043 0.08307 0.03987
Proportion of Variance 0.00091 0.00081 0.0006 0.00052 0.00027 0.00023 0.00005
Cumulative Proportion  0.99749 0.99830 0.9989 0.99942 0.99969 0.99992 0.99997
                          PC29    PC30
Standard deviation     0.02736 0.01153
Proportion of Variance 0.00002 0.00000
Cumulative Proportion  1.00000 1.00000
```

> Q4. From your results, what proportion of the original variance is captured by the first principal component (PC1)?

```r
summary(wisc.pr)$importance[2, 1]
```

```
[1] 0.44272
```

> Q5. How many principal components (PCs) are required to describe at least 70% of the original variance in the data?

```r
which(summary(wisc.pr)$importance[3, ] >= 0.70)[1]
```

```
PC3
  3
```

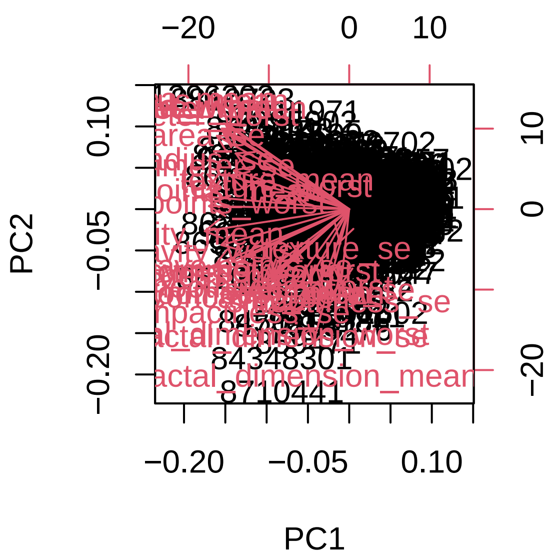
> Q6. How many principal components (PCs) are required to describe at least 90% of the original variance in the data?

```r
which(summary(wisc.pr)$importance[3, ] >= 0.90)[1]
```

```
PC7
  7
```

> Q7. What stands out to you about this plot? Is it easy or difficult to understand? Why?

```r
biplot(wisc.pr)
```



It is difficult to understand because all points are plotted together with different categorical names labeled.

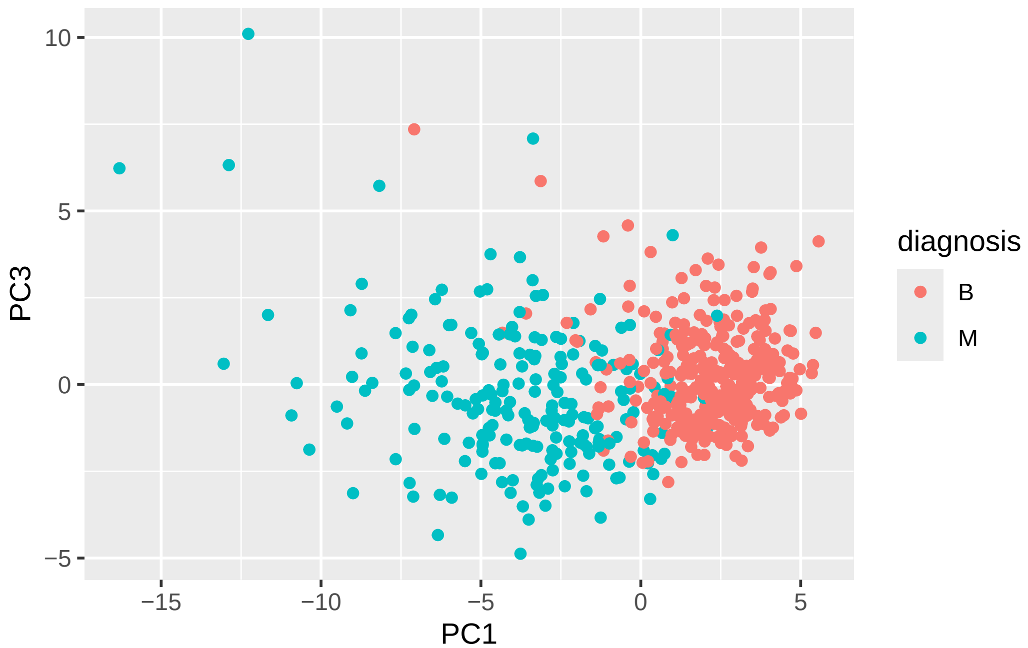
Q8. Generate a similar plot for principal components 1 and 3. What do you notice about these plots?

B and M seems to be seperated.

Let's see the "PC score plot"

```r
library(ggplot2)

ggplot(wisc.pr$x) +
  aes(PC1, PC3, col=diagnosis) +
  geom_point()
```



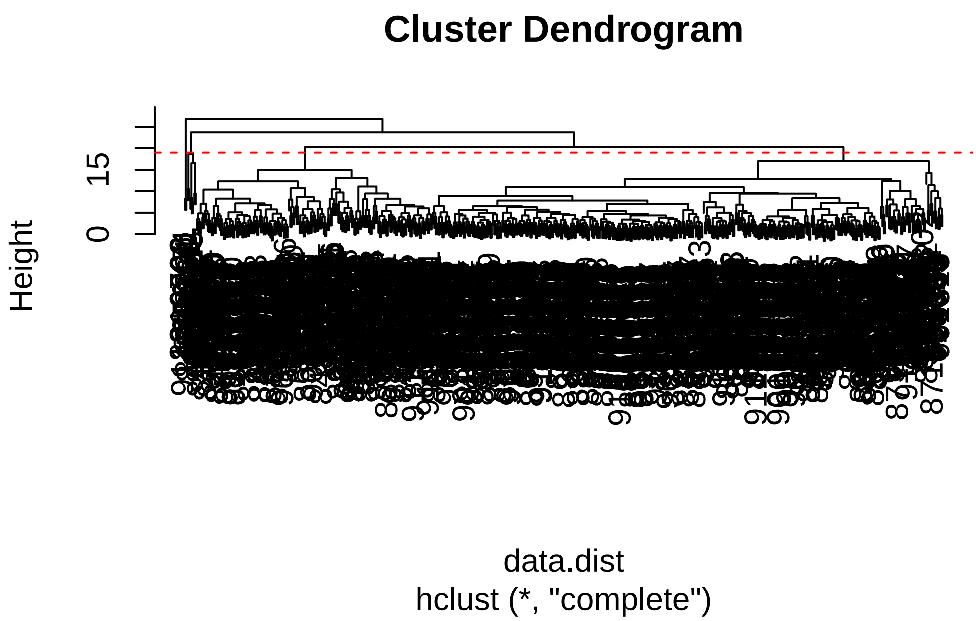
Q9. For the first principal component, what is the component of the loading vector (i.e. wisc.pr$rotation[,1]) for the feature concave.points_mean? This tells us how much this original feature contributes to the first PC. Are there any features with larger contributions than this one?

For the first principal component (PC1) in the Wisconsin breast cancer dataset, the loading for concave.points_mean is large and positive.This means it is one of the strongest contributors to PC1. There may be a few features with similar or slightly larger loadings but overall concave.points_mean is among the top contributors to the first principal component.

Hierarchical clustering

```
# Scale the wisc.data data using the "scale()" function
data.scaled <- scale(wisc.data)
data.dist <- dist(data.scaled)
wisc.hclust <- hclust(data.dist, method="complete")
```

```
plot(wisc.hclust)
abline(h=19, col="red", lty=2)
```



Q10. Using the plot() and abline() functions, what is the height at which the clustering model has 4 clusters?

h=20

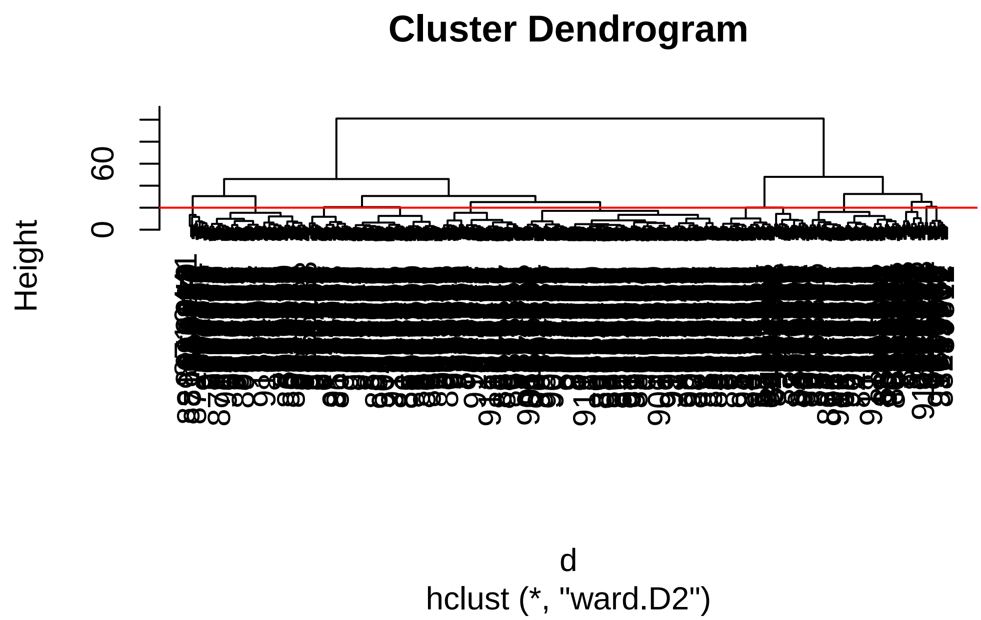
Combining methods

Q12. Which method gives your favorite results for the same data.dist dataset? Explain your reasoning.

Ward.D2 gives the best results.

It tends to produce compact, well-separated clusters because it minimizes within-cluster variance at each step. In contrast, single linkage often creates "chaining" effects, and complete/average linkage can be less balanced. Overall, Ward.D2 usually yields the most interpretable and meaningful clustering structure for this dataset.

```
d <- dist(wisc.pr$x[,1:4])
wisc.pr.hclust <- hclust(d, method="ward.D2")
plot(wisc.pr.hclust)
abline(h=20, col="red")
```




```
grps <- cutree(wisc.pr.hclust, h=70)
table(grps)
```

```
grps
  1   2
171 398
```

> Q13. How well does the newly created hclust model with two clusters separate out the two "M" and "B" diagnoses?

How does this clustering `grps` correspond to the expert `diagnosis`?

```r
table(diagnosis, grps)
```

```
         grps
diagnosis   1   2
        B   6 351
        M 165  47
```

The clustering corresponds fairly well to the expert diagnosis. Group 1 is mostly malignant: 165 M and only 6 B. Group 2 is mostly benign: 351 B and 47 M. So the clusters separate benign and malignant cases pretty well, though not perfectly because 53 total cases are placed in the opposite-majority cluster.

> Q14. How well do the hierarchical clustering models you created in the previous sections (i.e. without first doing PCA) do in terms of separating the diagnoses? Again, use the table() function to compare the output of each model (wisc.hclust.clusters and wisc.pr.hclust.clusters) with the vector containing the actual diagnoses.

```r
wisc.hclust.clusters <- cutree(wisc.hclust, k = 4)
table(wisc.hclust.clusters, diagnosis)
```

```
                    diagnosis
wisc.hclust.clusters   B   M
                   1  12 165
                   2   2   5
                   3 343  40
                   4   0   2
```

The hierarchical clustering without PCA does a poorer job separating the diagnoses. The clusters contain a more mixed combination of benign and malignant cases, indicating less clear structure in the data. Overall, compared to the PCA-based clustering, it is less effective at distinguishing between the two groups.

> Q16. Which of these new patients should we prioritize for follow up based on your results?

We should prioritize patient 2 for follow-up.

From the PCA plot, patient 2 lies closer to the cluster associated with malignant (M) cases, while patient 1 is closer to the benign cluster. This suggests patient 2 is at higher risk and should be examined more urgently.